

TORSVAG FYR, Norway

WMO: 010330

Lat: 70.246N

Lon: 19.500E

Elev: 79

StdP: 14.65

Time Zone: 1.00 (EUC)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	19.7	21.8	6.3	7.6	22.8	9.6	9.0	24.0	44.4	32.2	40.6	32.4	17.6	120	0.802

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.6	65.7	56.3	62.8	54.6	60.2	53.4	57.6	64.0	55.7	61.1	54.0	58.9	9.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
54.3	62.8	60.5	52.5	58.9	58.2	51.2	56.0	56.4	24.9	64.2	23.7	61.4	22.7	59.3	64.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 36.8	32.1	28.2	DB	16.8	70.5	3.4	3.1	14.3	72.8	12.3	74.6	10.4	76.4	7.9	78.6	(4)
(5)			WB	14.0	60.1	3.6	2.1	11.4	61.6	9.3	62.8	7.3	64.0	4.7	65.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	40.8	31.9	31.4	32.1	36.5	42.6	47.3	52.4	52.1	48.6	42.1	37.2	34.3
	DBStd	9.08	5.29	5.20	4.91	5.25	5.48	4.75	4.80	4.43	5.01	5.21	4.95	5.15
	HDD50	3641	560	520	554	405	238	110	26	26	83	250	383	486
	HDD65	8842	1025	940	1019	854	693	532	392	401	491	711	833	951
	CDD50	276	0	0	0	1	10	28	100	90	42	5	0	0
	CDD65	1	0	0	0	0	0	0	1	0	0	0	0	0
	CDH74	2	0	0	0	0	0	1	1	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	12.5	15.3	15.7	15.5	12.0	10.9	9.7	8.1	8.4	11.3	13.2	14.4	16.1
Precipitation	PrecAvg	32.3	2.9	2.4	3.0	2.1	2.0	1.7	1.9	2.3	3.4	4.3	3.0	3.3
	PrecMax	40.3	6.9	4.7	5.9	5.8	5.4	4.8	5.3	5.3	6.6	8.7	6.1	5.5
	PrecMin	23.2	0.0	0.6	0.7	0.4	0.4	0.6	0.1	0.4	0.5	0.6	0.0	0.0
	PrecStd	4.4	1.8	1.0	1.3	1.1	1.2	0.9	1.3	1.3	1.6	1.7	1.7	1.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.2	44.0	45.5	54.1	62.6	66.2	71.5	67.5	63.3	57.3	50.3	46.9
		MCWB	40.5	39.5	38.2	44.7	50.8	55.5	59.2	57.4	53.4	49.5	44.6	42.0
	2%	DB	42.6	42.3	42.9	49.1	57.7	61.7	67.5	64.4	60.7	53.8	48.1	45.0
		MCWB	38.4	38.2	38.5	42.6	48.6	52.8	57.6	56.1	52.7	47.7	43.8	41.1
	5%	DB	41.0	40.9	41.1	45.5	53.9	57.9	63.4	61.7	58.0	50.9	46.2	43.3
		MCWB	37.3	37.1	37.0	40.3	46.3	51.1	55.3	54.7	51.4	46.2	42.2	39.4
	10%	DB	39.1	38.9	39.1	43.4	50.6	54.5	60.3	59.0	55.3	49.0	44.3	41.5
		MCWB	35.7	35.6	35.6	39.0	44.3	49.2	54.4	53.4	49.9	45.1	40.5	37.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	41.4	41.1	40.4	45.5	52.0	57.2	61.0	59.7	54.8	50.3	47.0	43.7
		MCDB	43.7	43.4	42.8	52.8	60.6	63.7	70.1	65.6	60.5	55.6	49.1	45.8
	2%	WB	39.1	38.6	39.0	43.0	49.1	53.5	58.4	57.1	53.4	48.6	44.6	41.7
		MCDB	42.1	41.2	42.2	47.8	56.8	60.1	65.4	62.8	59.2	52.3	47.4	44.6
	5%	WB	37.5	37.1	37.4	41.3	47.0	51.5	56.5	55.3	52.0	47.0	42.4	39.7
		MCDB	40.4	40.0	40.9	44.9	53.1	56.9	62.5	60.3	56.5	50.3	45.6	43.1
	10%	WB	35.9	35.6	35.7	39.6	44.9	49.5	54.5	53.8	50.7	45.4	40.6	37.7
		MCDB	38.9	38.7	38.8	42.9	49.3	53.8	59.3	58.1	54.6	48.8	43.9	41.0
Mean Daily Temperature Range	5% DB	MDBR	4.9	5.1	5.1	6.1	7.1	6.8	7.6	6.6	5.9	4.8	4.7	5.0
		MCDBR	6.7	6.8	6.3	8.4	12.4	13.0	14.5	11.7	9.6	6.7	6.9	7.1
		MCWBR	6.0	5.7	5.1	5.3	7.0	7.3	7.9	6.4	5.1	4.9	5.7	6.1
	5% WB	MCDBR	6.2	6.8	6.3	8.2	11.8	11.8	13.5	10.6	8.7	6.4	6.6	7.1
		MCWBR	5.9	6.3	5.7	5.5	6.9	7.1	7.9	6.2	5.3	5.3	6.0	6.6
Clear-Sky Solar Irradiance	taub	N/A	0.271	0.322	0.335	0.344	0.341	0.346	0.335	0.317	0.272	N/A	N/A	
	taud	N/A	2.097	2.199	2.334	2.438	2.498	2.516	2.529	2.522	2.188	N/A	N/A	
	Ebn at Noon	0	134	210	252	266	271	265	254	224	132	0	0	
	Edh at Noon	0	12	23	28	29	28	27	23	17	11	0	0	
All-Sky Solar Radiation	RadAvg	0	71	372	928	1418	1550	1352	953	505	144	0	0	
	RadStd	0	12	43	95	108	163	144	72	56	22	0	0	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (17 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page